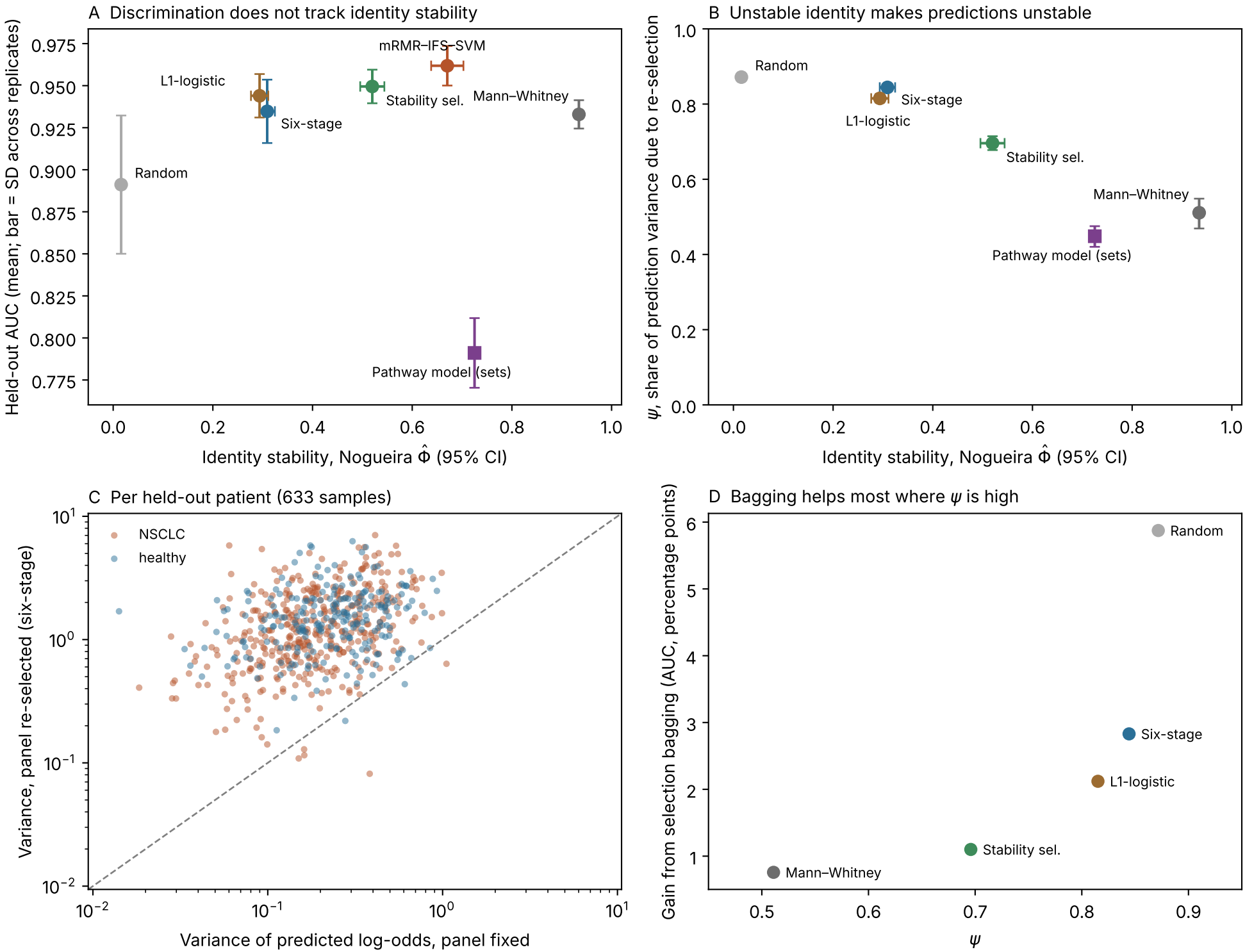
A Discrimination does not track identity stability
Held-out AUC (mean; bar = SD across replicates)
Identity stability, Nogueira Φ̂ (95% CI)
mRMR–IFS–SVM
L1-logistic
Stability sel.
Mann–Whitney
Six-stage
Random
Pathway model (sets)
B Unstable identity makes predictions unstable
ψ, share of prediction variance due to re-selection
Identity stability, Nogueira Φ̂ (95% CI)
Random
Six-stage
L1-logistic
Stability sel.
Mann–Whitney
Pathway model (sets)
C Per held-out patient (633 samples)
Variance, panel re-selected (six-stage)
Variance of predicted log-odds, panel fixed
NSCLC
healthy
D Bagging helps most where ψ is high
Gain from selection bagging (AUC, percentage points)
ψ
Random
Six-stage
L1-logistic
Stability sel.
Mann–Whitney